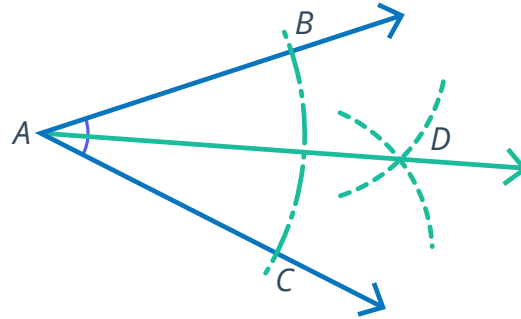


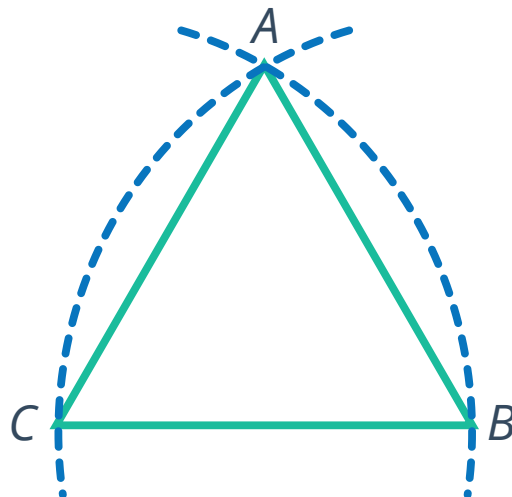
Justifying constructions using congruent triangles

- 1 Consider the construction that has been created so that the arcs with centers at B and C are congruent. Describe $\overrightarrow{AD}$.

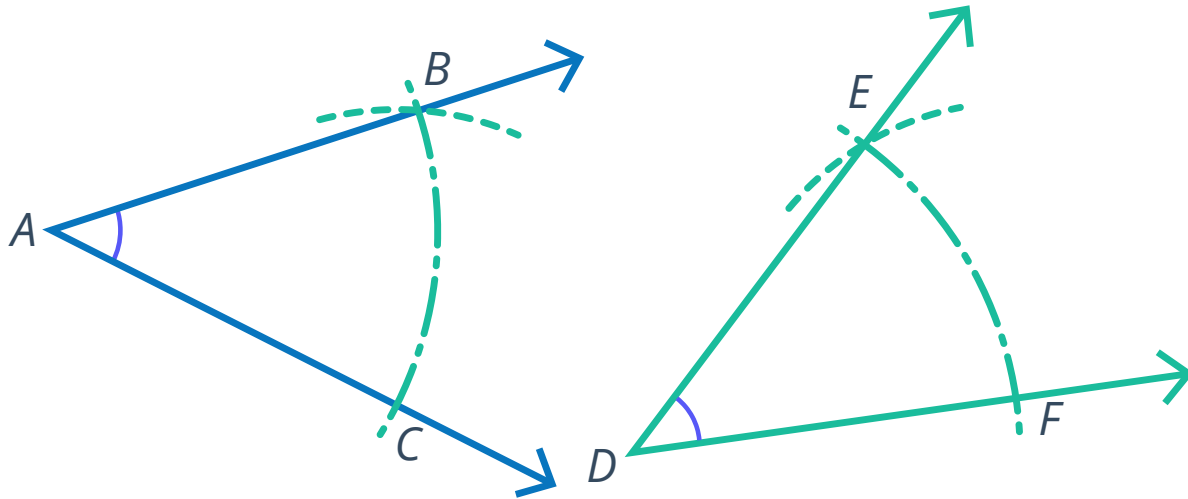


- 2 Consider the arc of the circle centred at C that passes through B and the arc of the circle centred at B that passes through C . Both arcs intersect at a point A , and the triangle ABC is constructed as shown:

- a Write line segments that are congruent radii of the circle centred at C .
- b Write the line segments that are congruent radii of the circle centred at B .
- c Classify the triangle ABC as scalene or equilateral.



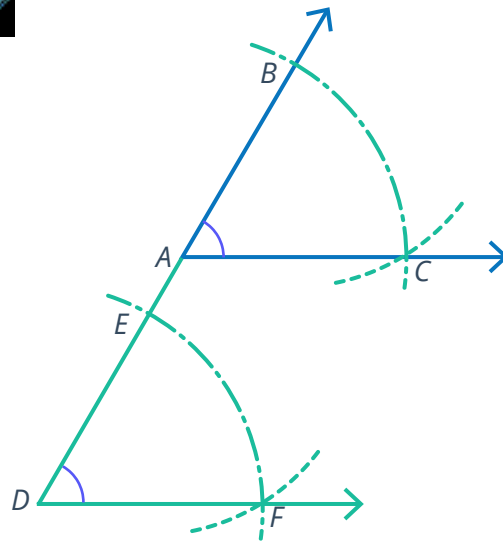
- 3 Consider the angle $\angle EDF$, constructed from  given angle $\angle BAC$ by drawing a set of dotted arcs.



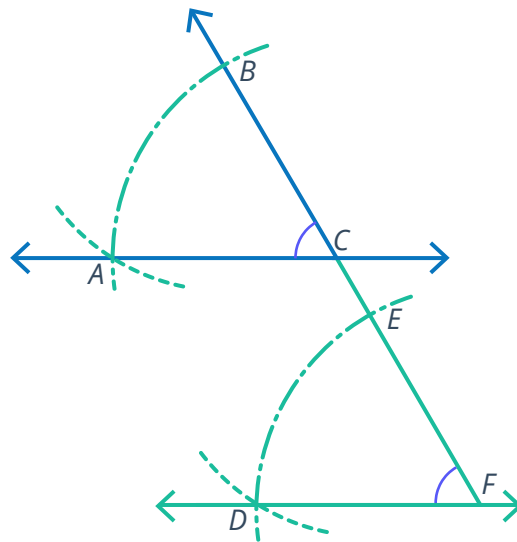
It can be determined from the construction that:

- $\overline{AB} \cong \overline{DE}$
 - $\overline{AC} \cong \overline{DF}$
 - $\overline{BC} \cong \overline{EF}$
- a State the reason that the line segments above are congruent.
- b Are the triangles $\triangle ABC$ and $\triangle DEF$ congruent? If yes, state the congruence theorem that justifies the congruence.
- c Determine whether the following reasons prove that $\angle BAC$ and $\angle EDF$ are congruent.
- i It can be shown that the angles are corresponding parts of congruent triangles.
 - ii The angles were measured with a protractor and shown to have approximately the same measure.
 - iii The angles look like they are the same size.

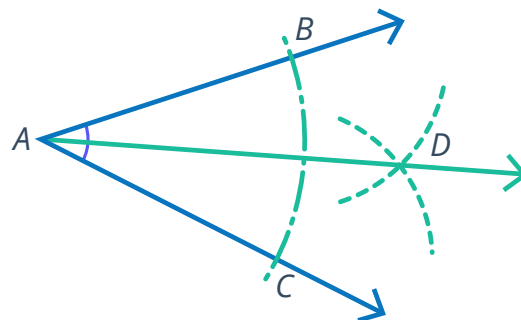
- 4 Consider the given diagram, where $\angle EDF$ is constructed congruent to $\angle BAC$. Justify the steps of construction in a two-column proof.



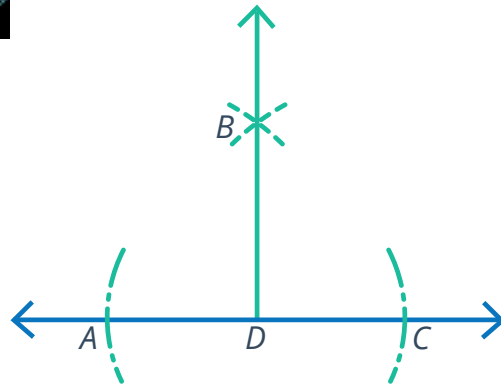
- 5 Consider the given diagram, where $\overleftrightarrow{DF}$ is constructed parallel to $\overleftrightarrow{AC}$. Justify the steps of construction in a two-column proof.



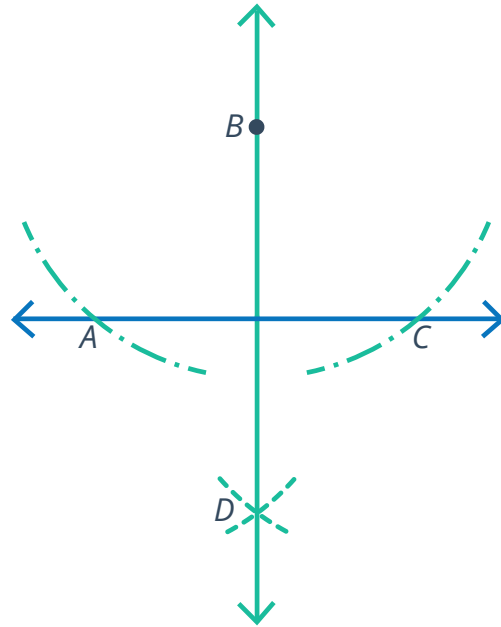
- 6 Consider the given diagram:
Prove that the construction of $\overrightarrow{AD}$ is an angle bisector of the given angle $\angle BAC$.



- 7 Consider the given diagram, where $\overleftrightarrow{DB}$ is constructed perpendicular to $\overleftrightarrow{AC}$. Justify the steps of construction in a two-column proof.



- 8 Consider the given diagram, where $\overleftrightarrow{BD}$ is constructed perpendicular to $\overleftrightarrow{AC}$. Justify the steps of construction in a two-column proof.



- 9 Consider the given diagram.

- a $\overleftrightarrow{CD}$ is constructed from the given line $\overleftrightarrow{AB}$ by drawing a set of dotted arcs. Prove that the point E is the midpoint of $\overleftrightarrow{AB}$.
- b $\overleftrightarrow{CD}$ is constructed so that it is a perpendicular bisector of $\overleftrightarrow{AB}$ and $\overleftrightarrow{AB}$ is a perpendicular bisector of $\overleftrightarrow{CD}$. Justify the steps of construction in a two-column proof.

